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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
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American Chemical Society

Dear Prof. Merz,

We submit our manuscript **"Resistance-Aware Polypharmacology of Antimalarial Leads from African Natural Products: Docking, Network Scoring, and Targeted Molecular Dynamics"** for consideration as an Article in *JCIM*.

This work tests a layered, resistance-aware prioritisation strategy rather than assigning a systems-level mechanism of action from docking alone. The 17 Set-C candidates were evaluated by docking-derived RRS across six resistance states, with ACSI and PNS providing complementary chemical-space and network-ranking dimensions; four separate parent-study complexes underwent targeted wild-type MD (10 ns each; 40 ns total). Six candidates were Class A*, five Class B, five Class C, and one Class D under the per-target RRS definition. Only PfCRT–214 yielded an interpretable MM-GBSA estimate; the other parent-study systems were dissociated or conversion-excluded. Tartarus analysis provided no evidence of a composite MPO–affinity association, and this null result is interpreted as separation of ranking dimensions rather than proof of statistical independence.

The code, compact scoring outputs, and source-data manifests are available in the GitHub repository under an open-source licence. Large parent-study trajectories are identified with their analysis boundaries and are not used beyond the systems and checksums that can be verified. Statistical interpretation is bounded by effect sizes, exact p-values, and pre-specified Bonferroni correction; incomplete Set-C MD-RRS is not presented as a result.

Suggested reviewers.

1. Dr. Fidele Ntie-Kang, University of Buea — African NP computational discovery
2. Dr. Kelly Chibale, University of Cape Town — antimalarial drug discovery
3. Dr. John D. Chodera, Memorial Sloan Kettering — free energy methods, MD best practices

This manuscript is original, not under consideration elsewhere, and all authors have approved the submission.

Sincerely,

Myke-Vital Temgoua Sao
(on behalf of all co-authors)